

AI Workshop #0

Course Introduction

1. Course Objectives

As an integration of the AI2013 Introduction to Artificial Intelligence course which focuses on the basic concepts and knowledge of AI with the basic Python programming skill learnt in AI1003 Python Programming, this AI workshop will focus on how various key AI technologies work for the implementation of AI applications using Python, which include:

- Machine Learning
- Logic Programming
- Heuristic Search Techniques
- AI Games
- Artificial Neural Networks
- Deep Learning and Deep Networks

2. Course Lecturer/TA

Course Lecturer

Dr. Raymond LEE (1001)

Prof. Weifeng SU (1002)

Course TAs

Ms. Rubin ZHOU (1001)

Mr. Runlin HUANG (1002)

3. Teaching Mode & Approach

Teaching Mode

- Workshops base
- 3-hour lab workshop per week

Teaching Approach

- Python JupyterNB teaching with the combination of lectures + labs

4. Course Content

1. AI with Python Primer Concepts
2. A Revisit on Machine Learning
3. Data Preparation with Python
4. Supervised Learning with Classification
5. Predictive Analytics with Ensemble Learning
6. Detecting Patterns with Unsupervised Learning
7. KNN for Building Recommendation Systems
8. Logic Programming
9. Heuristic Search Techniques
10. AI Games
11. Artificial Neural Networks
12. Deep Learning with Convolutional Neural Networks
13. Recurrent Neural Networks (RNN)

5. Assessment Methods

Type of Assessment	Weighting	Description of Assessment Tasks
Workshop Assignments	20%	Individual Workshop Assignments measure students' ability to use the technologies taught in this course. It will also test and reward students' problem-solving capabilities and coding styles. (With 5% as Attendance)
Group Term Paper	25%	Term paper to assess the students' capability of literature review of key AI technology and the training of their technical writing skill.
Quizzes	15%	The quizzes to assess the major learning outcomes achieved by the students.
Group Project	40%	Term program project (with presentation) to assess students' capability by applying the concepts and skills learnt from the course to implement AI applications.

6. Recommended Readings

1. R. Lee. Artificial Intelligence in Daily Life, Springer, 2020.
2. Russell, S. & Norvig, P. Artificial Intelligence: A Modern Approach, Prentice-Hall, 2020.
3. Philip C. Jackson, Introduction to Artificial Intelligence: Third Edition. 2019
4. Wolfgang Ertel, Introduction to Artificial Intelligence 2nd ed. 2017.
5. Russell and Norvig, A Modern Approach, Prentice-Hall, third edition. 2010
6. Bonaccorso, G. Mastering machine learning algorithms. Packt Publishing, 2018.
7. Rothman, D. Artificial intelligence by example. Packt Publishing, 2018.
8. Artasanchez, A., & Joshi, P. Artificial intelligence with python. Packt Publishing, 2020.
9. S. Marsland. Machine Learning: An Algorithmic Perspective, 2nd edition. Chapman & Hall/CRC, 2015.
10. Vasilev, I. Advanced deep learning with python: Design and implement advanced next-generation AI solutions using TensorFlow and pytorch. Packt Publishing, Limited, 2019.



Group Project: Building Your AI Chess Bot

So far, we have learnt:

- Different basic types of AI-based search algorithms for winning in a chess game such as Combinational search, Minimax algorithm, Alpha-beta pruning, Negamax algorithm, SSS* algorithm, etc.
- Implementation of various basic AI chess games such as Tic-Tac-Toe, Connect-FOUR and Hexapawn.

In the Group Project, form a group to design and implement an AI-based Chess Bot with:

- At least 3 levels of AI challenges, one above the others
- Build both interactive H2M and M2M - non-interactive AI Chess Game to demonstrate Chess Bot in various levels of challenge.
- Highly recommend to add your own AI algorithm to beat the classical methods
- Extra marks will be given with nice looking interface.

Deadline and Presentation: Week 14



Project Term Paper (Technical Report)

As part of the project deliverable, each project group have to submit a Project Term Paper to discuss the literature review of AI-game/AI-Chess bot, related technology and their proposed AI Chess Bot with experimental results/ simulation results.

The TOC (Table of Content) include:

- Project Title
- Abstract (200 - 250 words)
- Keywords (4 - 6)
- Introduction
- Literature Review
- Methodology
- Implementation
- Experimental Results
- Performance Analysis
- Future Work
- Conclusion
- Reference

Note:

1. Total no. of page: 8 - 12 pages, double-column, single-spacing
2. Paper format: Please refer to the attached IEEE Paper Template in the directory folder.

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